Source: https://cam.feedzai.com/cs-hsbc/3.0.2/docs/installation-guide/dtp/service-dtp

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# DTP Service

The Data Transfer Pipeline service (DTP) aims to be a more robust and scalable solution in comparison with ETL, by solving the following limitations:

- Better support for workflow schema change through the use of a workflow service that has information about the last published schema
- Events are streamed as soon they are processed in Pulse and sent to an upstream cloud native service (GCP Pub/Sub)
- Historical events extracted as they reach end of the Pulse workflow - it does not matter when they happened
- All the event fields that reach the end of the workflow will be sent to the Data Transfer Pipeline
- Data is sent in near real time to DSE and the extractions contain the metric values so the data sources can be immediately used (if there are not new metrics in the plan under test)

In this page one can find information about the service architecture, usage and monitoring.

## Architectureâ

The DTP service uses the following GCP resources:

- Pub/Sub topics
- Pub/Sub schema registry
- Dataflow pipelines
- GCS buckets

Please refer to the DTP permissions page for the permissions required by the service account attached to the machine running the Data Transfer Pipeline (DTP).

The service architecture and the GCP resources utilization is represented in the following diagram:

### Data Generationâ

An instance of the service runs in each Pulse workflow, streaming data generated in the Pulse workflow to GCP Pub/Sub.

### Data Processingâ

The near real time messages streamed from the service are consumed by `1 + N` subscribers:

- DTP subscriber that consumes and transforms the data and makes it available to Pulse DS
    - This subscriber uses a Dataflow pipeline to process and store the data in a GCS bucket
- Any other subscribers developed by HSBC to consume and process messages. For instance, there can be a subscriber that processes and sends messages to FDR

### Data Consumptionâ

Data generated from the DTP subscriber is available to be used in Pulse DS for risk strategy tests and upgrades. Any data consumed/generated by other subscribers can be used in other system (e.g. FDR).

## Usageâ

As previously stated, DTP is a Pulse service and therefore has to be added to a Pulse application workflow.

By default it should appear in the Custom Services section in Pulse UI by going into the `Data Configuration` tab of an application. If it does not appear it can be manually added according to the following configuration:

```
Name: Data Transfer Pipeline
Type: Workflow
Binary: Provided
Binary Path: data-transfer-pipeline-service.jar
```

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If the service is loaded correctly a set of configurations will then appear in the `Configuration` section. Please refer to [DTP configuration reference page](#) to understand how to configure the service.

After the service is configured it needs to be added to the workflow. To do so:

- Go the application workflow page
- Select the workflow
- Add a new `Custom Workflow Element` to the workflow
- Add a name (e.g. DTP), leave the `Target Outcome` empty and select `Data Transfer Pipeline` in the `Custom Service` dropdown
- Save the workflow and publish

Publishing the application workflow with the DTP service enabled will trigger the creation/update of the associated GCP resources.

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The DTP service doesn't automatically remove any GCP resources, except the dataflow jobs marked as inactive by the Inactivity service.

## Inactivity Serviceâ

The DTP service has an embedded inactivity service that detects and cancels inactive Dataflow jobs.

When enabled:

- Each Dataflow job pipeline includes an extra step that monitors the job inactivity
- If inactivity is detected, the service sends a message to the designated inactivity Pub/Sub topic
- The pipeline is then canceled, provided it is not the only active pipeline for its associated Pulse application and cluster color

Please refer to the Inactivity service section in the configuration page to understand how to configure the service.

## Data Securityâ

The DTP service allows configuring security measures related to data encryption and regional restrictions.

### Data Encryptionâ

It's possible to specify a customer-managed encryption key (CMEK) to encrypt data handled by Pub/Sub topics and Dataflow pipeline resources using the following property:

- `security.kmsKeyName`

The value should follow the format described in Configure a topic with CMEK.

If omitted, Google-managed encryption keys will be used by default.

### Regional Restrictionsâ

It's possible to control where Pub/Sub is allowed to store and transmit data using the following properties:

- `security.allowedRegions` : A comma-separated list of allowed Google Cloud regions where Pub/Sub may store data. Example: "europe-west1,asia-east2".

```
# Ansible variable
dtp_security_allowed_regions:
  - europe-west1
  - asia-east2
```

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- `security.enforceInTransit` : A boolean flag that, when enabled, enforces in-transit data restrictions to the same regions defined in `security.allowedRegions` .

```
# Ansible variable
dtp_security_enforce_in_transit: true
```

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If regional restrictions are active, the following property becomes **mandatory**:

- `security.regionalEndpoint` : The regional Pub/Sub gRPC service endpoint to use when regional restrictions are enforced. Example: "europe-west1-pubsub.googleapis.com:443". The full list of available endpoints can be found [here].

```
# Ansible variable
dtp_security_regional_endpoint: "europe-west1-pubsub.googleapis.com:443"
```

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When no regional restrictions are in place, the global Pub/Sub endpoint is used. This endpoint automatically routes traffic to the closes region to minimize latency.

Therefore, when using a regional endpoint, it's important to choose the appropriate one - preferably in the same region as other GCP resources and Pulse. If the regional endpoint is in a different region, performance can be affected, as cross-region requests are typically slower.

## Logging and Monitoringâ

By default, the service has an `INFO` log level. This value can be changed by using the following Ansible variable:

```
data_transfer_pipeline_log_level: info
```

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To build a monitoring solution for the DTP service please refer to the DTP Metrics Guide.